

GHOST IN TRANSLATION

DS1

1.0.X

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1. INTRODUCTION

DS1 is a multi-algorithm Drum Synthesizer module based on my Motherboard platform.

Features

- 16 algorithms
- 32 bit float DSP
- 48kHz 32 bit audio
- MIDI TRS Type A IN and OUT
- Auto calibration
- Either-way power

Technical Specifications

- **Module Width**: 8 HP
- **Depth**: 45 mm
- **Power Consumption**:
  - **+12V**: 75 mA
  - **-12V**: 3.5 mA
- **CV Input Range**: 0V to 5V

2. INSTALLATION

Turn off and disconnect your Eurorack case from AC mains power before installing this module. Connect the supplied power ribbon cable to the module and your busboard. This module is equiped with either-way power, so no magic smoke will be released if you plug it in backwards.

3. CONTROLS AND CV INPUTS

All CV inputs read voltages between 0 and 5V.

Knobs

- **Pitch**: Sets the base pitch/tuning for the voice.
- **Decay**: Controls the envelope decay time.
- **Tone**: Controls the brightness of the sound.
- **Drive**: Controls the gain of the drive/waveshaper stage.
- **Param 1**: Controls the primary sound synthesis parameter.
- **Param 2**: Controls the secondary sound synthesis parameter.

CV Inputs

- **CV Pitch**: Controls the pitch of the voice.
- **CV Decay**: Controls the decay of the voice envelope.
- **CV Accent**: Controls the accent level of the voice.
- **CV Trigger**: Triggers the voice.
- **CV Param 1**: Controls the primary sound synthesis parameter.
- **CV Param 2**: Controls the secondary sound synthesis parameter.
- **CV Select**: Selects the voice, 5v divided by 16 algorithms, about 0.3v steps.

Navigation Button

- **SEL button**:
  - **Short Press**: Manual trigger.
  - **Long Press (1 second)**: Enters algorithm selection mode.

4. MENU & NAVIGATION

Normal Play Mode

In normal play mode:

- The **8 LEDs at the top** display real-time levels of the control inputs and the main audio output.

Algorithm Selection Mode

To change the active sound algorithm:

1. **Enter Menu**: Long press the **SEL button**. All 8 top LEDs will turn solid **Blue**, except for the active selection.
2. **Read Active Selection**: The active selection is shown by two blinking **Green** LEDs:
    - **Bank Indicator** (one of the first 4 LEDs)
    - **Algorithm Indicator** (one of the last 4 LEDs)
3. **Change Algorithm**: Tap the **SEL button** to cycle to the next algorithm.
4. **Save & Exit**: After **5 seconds** of inactivity, the module saves your choice and returns to play mode automatically.

5. THE 16 ALGORITHMS

The module contains 16 algorithms. Below is an overview of what each algorithm does and what the **Param 1** and **Param 2** knobs control.

Bank 1

#	Sound Description	Param 1 Control	Param 2 Control
1	**Kick 1**: Dual-operator warm FM kick with rumble style sub.	Rumble amount	Pitch Sweep Drop Depth
2	**Kick 2**: Deep sine kick with FM attack punch and pitch sweep.	FM Transient Punch Amount	Pitch Sweep Drop Depth
3	**Kick 3**: Aggressive square-wave kick.	Pitch Sweep Duration	Pitch Sweep Drop Depth
4	**Kick 4**: 909 style dual sine/triangle oscillator kick.	Pitch Sweep Duration	Pitch Drop Depth

Bank 2

#	Sound Description	Param 1 Control	Param 2 Control
1	**Hi-Hat**: 808 style and 909 sample combination.	Metallic Frequency Ratio & Spread	Fader from synthesized to sample
2	**Metallic cowbell**: Classic cowbell to metallic noise.	Wavefolder depth	Crossfading cowbell to metallic cymbal oscillators
3	**Shaker**: Physical shaker simulation.	Jitter speed	Grit and resonance
4	**Cymbal**: 808 style and 909 sample combination.	Metallic Pitch Sweep Range	Fader from synthesized to sample

Bank 3

#	Sound Description	Param 1 Control	Param 2 Control
1	**Clap**: Classic pulsed noise.	Pulses spacing	Tail decay
2	**Tokyo Bell**: Life is simple. You make choices and you don't look back.	Detuning	Harmonics level
3	**Rings**: Physical modeling similar to the famous Rings by MI.	Resonator Modal Structure & Spacing	Strike Position
4	**Tom**: 808 to 909 tom.	FM Index & Sweep	Noise level

Bank 4

#	Sound Description	Param 1 Control	Param 2 Control
1	**Gong**: 4-operator FM inharmonic metallic.	FM Modulation Index	Inharmonic Wash Index
2	**FM Synth**: Phase modulation voice.	Modulator Frequency Multiplier	Phase Modulation Index Depth
3	**Acid Bass**: 303 style bass.	Filter Envelope Sweep Depth	Filter Resonance
4	**Dual Osc**: Disconant synth voice.	Waveform	Detune Amount

7. MIDI IMPLEMENTATION

DS1 listens to and transmits on **Channel 1**.

Control Change (CC) Mapping

- **CC 1**: Pitch
- **CC 2**: Decay
- **CC 3**: Tone
- **CC 4**: Drive
- **CC 5**: Param 1
- **CC 6**: Param 2
- **CC 7**: Algorithm select

Notes Mapping

- **60 (C3)**: Algorithm 0
- **61 (C#3)**: Algorithm 1
- **62 (D3)**: Algorithm 2
- **63 (D#3)**: Algorithm 3
- **64 (E3)**: Algorithm 4
- **65 (F3)**: Algorithm 5
- **66 (F#3)**: Algorithm 6
- **67 (G3)**: Algorithm 7
- **68 (G#3)**: Algorithm 8
- **69 (A3)**: Algorithm 9
- **70 (A#3)**: Algorithm 10
- **71 (B3)**: Algorithm 11
- **72 (C4)**: Algorithm 12
- **73 (C#4)**: Algorithm 13
- **74 (D4)**: Algorithm 14
- **75 (D#4)**: Algorithm 15

8. FIRMWARE UPDATE

The firmware update process is as follows:

1. **Power Off the Module**: Disconnect the module from power.
2. **Press and hold BOOTSEL**: Locate the physical **BOOTSEL** button on the microcontroller (Pico 2) on the back of the module, and hold it down.
3. **Connect to Computer**: While holding the **BOOTSEL** button, connect the Pico 2 to your computer via a USB cable.
4. **Release BOOTSEL**: Once the USB cable is plugged in, release the button. The Pico 2 will mount on your computer as a USB mass storage drive.
5. **Transfer the Firmware**: Locate the compiled firmware file `firmware/ds1.ino.uf2`. Drag and drop (or copy) the `.uf2` file onto the drive.
6. **Automatic Reboot**: The Pico 2 will write the new firmware to its flash memory, automatically eject/unmount, and reboot with the updated firmware running.

9. TROUBLESHOOTING

No Sound Output

- Verify that the eurorack power ribbon cable is connected correctly.
- Make sure your audio cable is patched securely.
- Switch off and switch on the module.